

Language-4D Cross-Boosting for Generalized Zero-Shot 6DoF Tracking and 3D Reconstruction

Jingtao Sun, Yaonan Wang, Jiawen Zhao, Yike Zhang, Min Liu, and Mike Zheng Shou

Abstract—A more universal category-level pose tracking and reconstruction approach should be robust to both seen and unseen classes. However, the performance of existing category-level algorithms degrades for classes that are unseen in the training set. To learn a discriminative model that can achieve strong performance in Generalized Zero-Shot Learning (GZSL) settings, we introduce L4D-Track++, a unified framework for zero-shot category-level pose tracking and shape reconstruction that leverages multi-level language captions. To address the bias problem, namely the out-of-distribution gap between source (seen) and target (unseen) domains, we first propose a Multi-Level Language Conditional Feature Generation module to produce synthetic pairwise embeddings for unseen classes. Furthermore, to increase inter-class distance, decrease intra-class distance, and bridge the gap between seen and unseen category domains, we propose a language-guided multi-modal fusion strategy to construct the final fused inter-frame pairwise embeddings for downstream tracking and reconstruction tasks by taking both real and synthetic pairwise embeddings as inputs. This strategy consists of two main components: Category-Related Mutual Information-based Optimization and Modality-Related Conditional Mutual Information Maximization. Experimental results on five public datasets demonstrate that our method is effective and achieves significant improvements in GZSL performance.

Index Terms—Zero-shot learning, multi-modality embedding, category-level pose tracking and shape reconstruction

1 INTRODUCTION

SIX Degrees-of-Freedom (6DoF) object pose estimation or tracking and 3D shape reconstruction are two core topics in computer vision, robotics, and virtual reality. Given a sequence of RGB-D frames depicting an object, the goal of 6DoF tracking is to estimate the object’s relative pose change between consecutive frames, while 3D shape reconstruction aims to recover a complete and temporally consistent 3D shape of the object from partial observations. These tasks are often divided into two manners: instance-level and category-level approaches. Early approaches primarily focused on instance-level setups, which require precise CAD object models. However, obtaining accurate models is often difficult and time-consuming, limiting their practical applicability for real-world scenarios. In this regard, the category-level problem has gained increasing attention, as it can handle unseen instances within known categories without requiring CAD models. A significant challenge in category-level tasks is intra-class shape/texture variation within each category. Besides, object instance categories in the real world often follow a long-tail distribution. Current research relies on short-tailed distributions using imbalanced sets, where some categories have abundant training samples, while others may have few or even none. As a result, traditional methods struggle to handle samples from classes that are not covered in the training data.

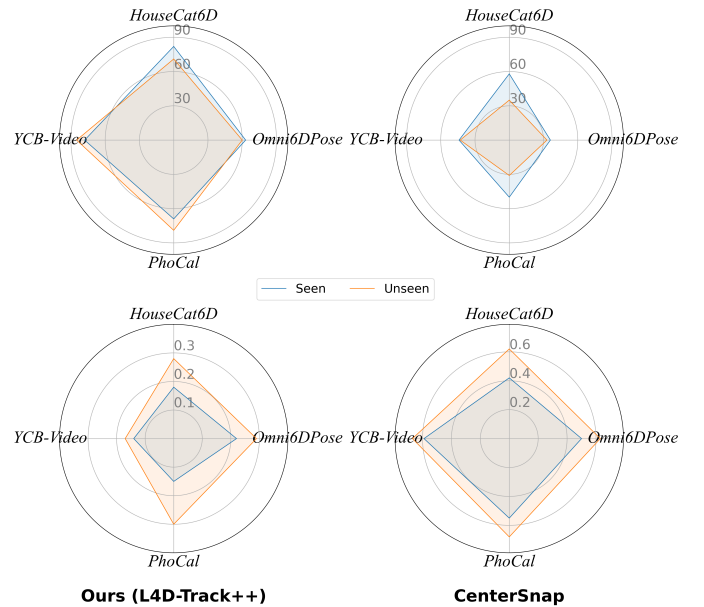


Fig. 1. Radar-map of GZSL performance for our L4D-Track++. We establish well GZSL performance for seen/unseen classes under category-level pose tracking (figure above with $mIoU$ metric $\uparrow$) and shape reconstruction (figure below with CD metric $\downarrow$) tasks on four unseen benchmarks. We also provide a comparison of CenterSnap.

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Project page is released at https://github.com/S-JingTao/L4D_Track
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Recently, Zero-Shot Learning (ZSL) techniques have made significant advancements, providing promising solutions to these long-tail issues. Conventional ZSL aims to train models capable of classifying objects from unseen classes (target domain) by transferring knowledge from seen classes (source domain), utilizing semantic information. However, this approach often relies on an unrealistic setting where the test dataset contains only unseen class samples,

which does not reflect real-world conditions. Due to the ubiquitous demand for machine learning techniques for large-scale multiple datasets, conventional ZSL is commonly applied to solve problems related to object recognition [1], classification [2], [3] and other popular applications, such as video processing [4] and natural language processing [5]. In this paper, we propose to develop a new research gap *i.e.*, zero-shot category-level 6-DoF pose tracking and 3D shape reconstruction task, integrated into the universal framework of Generalized Zero-Shot Learning (GZSL) [6]. In our main task, the objective is to predict (1) the inter-frame 6DoF pose transformations and (2) a complete 3D shape reconstruction of the object across time. This framework can address the challenge of handling samples from both seen and unseen classes simultaneously. Unlike traditional ZSL methods that suffer from a very serious strong-bias problem due to overfitting on seen classes, current GZSL-based solutions generally follow feature generation paradigms [2], [3], [7], [8] that leverage the synthesized unseen features conditional on unseen semantic embeddings to achieve GZSL in a supervised way. However, they typically generate visual embeddings within the image feature space without cross-linking them to the semantic space, which limits generalization capabilities. Meanwhile, they do not deal with the appearance variance of different domains, hence perform poorly in the generalized setting.

The different data distribution between seen and unseen classes causes difficulties in transferring knowledge from the seen classes to unseen classes (known as the bias problem), which is a key challenge for GZSL tasks. To alleviate this issue, many generative-based GZSL methods enhance the semantic information using textual descriptions [9] or large language models [10], *etc.* [11], [12]. However, these approaches have a common limitation: they rely on visual-semantic feature modeling from seen classes, which is suboptimal for unseen classes, therefore, inadequate for our proposed GZSL task. Furthermore, shape/texture variations in different instances of the same class cause extra challenges in learning the association between the visual and semantic space, which in turn affect pose tracking and shape reconstruction performance for the same class. Besides, category variability between different instances of different classes is also a crucial issue that hinders knowledge transfer from seen to unseen classes and creates a challenge for out-of-distribution category-level representation and ultimately causes a degradation of the GZSL performance. Intuitively, we aim to tackle the GZSL-based challenges of category-level 6-DoF pose tracking and 3D shape reconstruction in this work. To achieve this goal, we need to address three main challenges:

- 1) *"How to obtain effective information that can solve the bias problem?"*. Since unseen samples are not available during the training phase, our GZSL model often faces difficulty in extracting distinguishable pose information and shape features for unseen samples;
- 2) *"How to increase inter-class distance and decrease intra-class distance at the same time to boost 4D visual-language association for our proposed GZSL task?"*. It is inevitable that common objects have shape/texture variations, as well as category variability in the real-

world scene, and hence it is desirable to enhance our proposed GZSL model to effectively handle different instances within diverse categories;

- 3) *"How can intra-modal and inter-modal feature fusion be achieved in data-hungry situations?"*. It is well known that, at present, there is a serious shortage of the huge amount of multi-modal data for 3D vision, disabling traditional GZSL-based methods from being directly applied to our proposed task with small-scale data. Annotating such large-scale multimodal 3D datasets is both expensive and time-consuming. Hence, a natural idea is to adopt a modality fusion strategy, which can effectively work on small-scale data.

In the end, this work aims to resolve the above challenges by developing an effective and efficient GZSL-based category-level 6D tracking and 3D reconstruction framework named L4D-Track++, which builds upon our previous work, L4D-Track [13]. To address the bias problem, we leverage multi-level language captions to generate synthetic pairwise embeddings for unseen classes, using these synthetic features to mitigate the imbalance between seen and unseen classes. The proposed Language-4D Cross-Boosting Fusion module is used to further enhance the consistency between the language space and the inter-frame embedding space and then strengthen the discriminability of the final fused inter-frame pairwise embeddings. This module consists of two main collaborative parts: Category-Related Mutual Information (MI)-based Optimization and Modality-Related Conditional Mutual Information (CMI) Maximization. Concretely, the Category-Related MI-based Optimization increases inter-class distance and decreases intra-class distance, facilitating better knowledge transfer from seen to unseen classes and removing irrelevant information for downstream tasks. It leverages variational mutual information bounds among the real/synthetic pairwise embeddings. The Modality-Related CMI Maximization aims to improve our GZSL performance by maximizing the CMI lower bound between these real/synthetic pairwise embeddings, conditional on language descriptions. Comprehensive experiments show that L4D-Track++ surpasses all prior works. Compared with other category-level approaches, our main contributions are:

- We introduce a novel framework that combines multi-level language captions and cross-boosting multi-modal fusion for GZSL-based category-level 6-DoF pose tracking and 3D shape reconstruction.
- We propose a multi-level language conditional feature generation, which incorporates synthetic pairwise embeddings for unseen classes using language descriptions to handle the bias problem.
- We present Category-Related MI-based Optimization and Modality-Related CMI Maximization modules, which leverage mutual information theory to improve GZSL performance by enhancing the distinguishability between seen and unseen classes.
- We evaluate our GZSL-based method on five public benchmark datasets and demonstrate significant improvements over prior work, as shown in Fig. 1.

This paper mainly expands upon our conference work published at CVPR2024 [13]. Overall, this paper presents

significant extensions in the following three aspects: (1) We integrate extra synthetic features from multi-level language captions into the proposed model to handle the bias problem between seen and unseen classes; (2) We propose a language-4D Cross-Boosting Fusion to improve the GZSL performance of the proposed model via applying the MI tractable variational bounds for both category-related and modality-related aspects to optimize unimodal and multimodal fusion; (3) We update the evaluation datasets, including PhoCAL [14], HouseCat6D [15] and Omni6DPose [16], and provide additional qualitative and quantitative results to showcase the GZSL performance of our method. The paper is structured as follows: Section 2 provides a brief review of related works. Section 3 and Section 4 introduce preliminaries and the proposed method. Extensive experimental results are presented in Section 5, followed by a conclusion in Section 6.

2 RELATED WORK

2.1 Conventional Zero-Shot Learning

In terms of the conventional zero-shot learning problem, the testing set only contains unseen classes and the seen classes are not considered. ZSL [17], [18], [19], [20] aims to transfer the mapping relationships between visual features and semantic attributes from seen to unseen classes using semantic information. This semantic information can be defined manually as attribute vectors [21], automatically extracted word vectors [22], context-based embeddings [23], or their combinations. Existing ZSL methods can be broadly divided into image-text embedding-based ZSL [24], [25], [26], [27] and attribute/object learning-based ones [28], [29]. The former involves the mapping of image and text features into a shared embedding space to enable compatibility learning. Chen *et al.* [19] proposed discovering object part aggregation to reduce virtual connections. The latter group focuses on separate learning of attributes and objects. Misra *et al.* [29] suggested learning transformations on individual classifiers for attributes and objects, while Nagarajan *et al.* [28] represented compositions as transformations of attribute matrices acting on object word vectors. Ma *et al.* [20] combined the advantages of these two approaches, focusing on learning both attributes and objects while emphasizing compatibility. However, these methods fail to account for the relationships between different domains, and therefore do not construct domain-discriminative representations within a unified framework.

2.2 Generalized Zero-Shot Learning

Socher *et al.* [22] were the first to introduce the concept of GZSL. Unlike traditional ZSL, GZSL requires the recognition of samples from both seen and unseen classes during inference. Existing GZSL techniques can be roughly grouped into two types: embedding-based [2], [30], [31], [32], [33] and generative-based approaches [8], [34], [35], [36], [37], [38]. Embedding-based methods either enable knowledge transfer via aligning a common embedding space [2] or by utilizing attribute descriptions to discover discriminative local features [30], [32], [33]. Feinglass *et al.* [30] employed a pre-trained Vision-Language detector to

obtain region-specific attribute features. Li *et al.* introduced the idea of DSECN [31], which explores diverse semantics from external class names. In order to alleviate the bias problem, most of these generative-based methods have focused on the unrealistic setting where unseen classes are assumed to be similar to seen classes, resulting in poor performance. Wang *et al.* established an end-to-end framework called D³GZSL [34], which comprises in-distribution dual space distillation and out-of-distribution batch distillation modules. Ge *et al.* [35] developed DENet, a generative model that adapts training features and imposes constraints on generated features. While we also follow the generative-based pipeline, we utilize multi-level language captions to synthesize features to bridge the seen and unseen domains, leading to more effective modeling of the unseen domain distribution.

2.3 Category-Level Pose Tracking and Reconstruction

Early efforts in category-level object pose estimation aimed to predict 6-DoF pose within the same category from a single frame. These works typically learn a category-specific canonical representation (*e.g.*, NOCS [39], UDA-COPE [40], and HS-Pose [41]), or model shape prior deformation (*e.g.*, SGPA [42] and SPD [43]). A natural extension of single-frame pose estimation is real-time pose tracking across continuous frames. Category-level pose tracking [44], [45], [46], [47], [48], [49], [50], [51], [51] is more challenging than instance-level tracking, due to the lack of supervision from known CAD models. Wang *et al.* [44] pioneered the use of anchor-based keypoint matching for tracking with 6-PACK. In CAPTRA [45], Weng *et al.* leveraged pose-canonicalized point clouds to track both rigid and articulated object poses. Sun *et al.* introduced ICK-Track [46], which addresses tracking by utilizing inter-frame consistent keypoints. Liu *et al.* [51] introduced KPA-Tracker to tackle the problem of articulated object per-part 6D pose tracking. While these keypoint-based methods are successful, the zero-shot category-level pose tracking problem remains largely unexplored.

Tracking and reconstructing rigid objects is an important area in computer vision and robotics. Many efforts have focused on free-form tracking without any geometric prior, such as DynamicFusion [52] and Deepdeform [53]. However, these methods are unable to reconstruct occluded objects, which distinguishes them from category-level estimation, which aims to rebuild the complete object configuration. Another way involves performing 4D reconstruction from RGB-D videos, such as Fusion4D [54] and 4DComplete [55], which capture the geometry both in space and time. Sun *et al.* [13] first proposed the zero-shot category-level pose tracking and reconstruction task, and Wen *et al.* [56] presented a Neural Object Field for 6-DoF tracking of an unknown object combined with neural 3D reconstruction. Unlike rigid tracking approaches, Xue *et al.* [57] presented GarmentTracking, a framework specifically designed for category-level garment pose tracking and reconstruction. While these methods show promising results, zero-shot long-tail category-level tracking and reconstruction remain underexplored, facing fundamental challenges such as the bias problem and category variability.

In contrast to the aforementioned methodologies, we consider the GZSL setting for simultaneous category-level pose tracking and shape reconstruction. Our proposed framework combines information from the language semantic space and 4D feature space to establish the transfer from the seen domain to the unseen domain. This allows for simple and efficient sharing of multi-modal information, enhancing our model's generalized performance in the GZSL setting. While existing language-conditioned methods [58], [59] only focus on category-level pose estimation in static settings, our work is the first to unify category-level pose tracking and 3D reconstruction under a GZSL framework.

3 PRELIMINARIES

3.1 Problem Statement

In the conventional ZSL task for category-level 6D tracking and 3D reconstruction proposed in [13], the goal is to learn to simultaneously track the 6-DoF pose and reconstruct 3D shape from unseen classes (U), where object categories were not seen during training. Instead, in the GZSL setting, we aim to train a model that can track/reconstruct samples from both seen (S) and unseen classes with $S \cap U = \emptyset$, making it more challenging for real-world scenes. Suppose $\{(p_i, y_i) | p_i \in P, y_i \in S\}$ denotes the seen class sample set, where p_i denotes spatial features from the point cloud stream $P = \{p^t\}$ and y_i is the corresponding class label. Likewise, the sample set from unseen classes is defined as $\{(p_i, y_i) | p_i \in P, y_i \in U\}$.

The objective of our proposed category-level GZSL task is to learn a categorical function $f_{GZSL} : P \mapsto S \cup U$, that can track pose $\mathcal{P} = \{T \in \mathbb{R}^3, R \in \mathbf{SO}(3)\}$ and reconstruct shape $\mathcal{S} \in \mathbb{R}^3$ of samples from both seen and unseen classes. The input of our model is RGB-D video streams along with per-frame object segmentation masks s_t , and per-frame point cloud p_t obtained via back-projection. The outputs are the relative 6-DoF pose change transformation $\Delta\mathcal{P} \in \mathbf{SE}(3)$ between consecutive frames, and a complete 3D shape $\mathcal{S} \in \mathbb{R}^3$ of the tracked object. Our model is trained using only labeled examples from the seen class, with no access to the corresponding CAD models.

3.2 Information Theory Background (MI)

Let x and y denote two random variables belonging to their corresponding realization spaces $x \in X$ and $y \in Y$, whose probability distribution functions are $p(x)$ and $p(y)$, respectively. Quantifying the degree of the relationship requires a metric that captures dependence among them; therefore, we focus on Mutual Information (MI), which is a reparameterization-invariant measure of dependency. MI is a concept from information theory that measures the dependency relationship between these pairs of variables, denoted by $I(X; Y)$,

$$\begin{aligned} I(X; Y) &= \mathbb{E}_{p(x,y)} \left[\log \frac{p(x,y)}{p(x)p(y)} \right] \\ &= \mathbb{E}_{p(x,y)} \left[\log \frac{p(x|y)}{p(x)} \right] = \mathbb{E}_{p(x,y)} \left[\log \frac{p(y|x)}{p(y)} \right], \end{aligned} \quad (1)$$

where $I(X; Y) \geq 0$, and it is also equivalent to the Kullback-Leibler divergence,

$$I(X; Y) = D_{KL}[p(x, y) || p(x)p(y)]. \quad (2)$$

4 METHODOLOGY

As depicted on the left side of Fig. 2, given the inter-frame pairs of observable inputs (p_{t-1}, p_t) from a 3D video, along with the segmented masks (s_{t-1}, s_t) of an arbitrary object of interest. The current mask s_t is obtained from an off-the-shelf segmentation model: SAM [60]. We first extract spatial features using a 3D backbone (PointNet [61]). Meanwhile, the masks are passed through a 2D backbone to generate image features, which are then back-projected and combined with the spatial features. It is important to note that the 3D and 2D backbones serve solely as feature extractors and are not updated during training (i.e., their weights are frozen); thus, we use pre-trained models from our pre-work [13]. The pre-trained 2D/3D backbones provide stable and semantically meaningful 2D–3D features, which is particularly important for the proposed mutual-information-based optimization. Therefore, we do not fine-tune these backbones and instead concentrate on improving cross-category generalization through language-guided modeling. Besides, we employ our proposed GPT-assisted association tool [13] to generate related multi-level language captions $X_C^{(V,S,O)}$, that are only used during training. We then feed the feature pairs $\bar{f}_{t-1}, \bar{f}_t$ into a cross-attention-based cross-coupled fusion layer to constructs real pairwise embeddings for the inter-frame features:

$$\begin{cases} \bar{f}_{t-1} = \sum_{i \in \Omega_t} \alpha((\Gamma(\bar{f}_{t-1}) + \Delta) \cdot \Gamma(\bar{f}_t^i) / \sqrt{K}) \cdot \Gamma(\bar{f}_t^i) \\ \bar{f}_t = \sum_{i \in \Omega_{t-1}} \alpha((\Gamma(\bar{f}_t) - \Delta) \cdot \Gamma(\bar{f}_{t-1}^i) / \sqrt{K}) \cdot \Gamma(\bar{f}_{t-1}^i) \\ \Delta = \text{linear}(\bar{f}_t - \bar{f}_{t-1}) \end{cases} \quad (3)$$

where Γ is the linear projection layer, α denotes the softmax operation, and Ω is the feature space. Δ represents the inter-frame point feature difference, which is used to offset data drifting between consecutive frames.

In general, we first process the 3D video stream into real pairwise embeddings for the currently seen classes using the method described above. These embeddings are then aligned with the corresponding multi-level language captions (see Sec. 4.1) only during training stage. We also generate the synthetic pairwise embeddings for unseen classes via these multi-level language captions (see Sec. 4.2). Our method leverages a cross-boosting fusion manner (see Sec. 4.2), which mainly involves two collaborative parts: 1) Category-Related MI-based Optimization, which enhances model discriminability using the MI variational bounds among the real and synthetic pairwise embeddings; 2) Modality-Related CMI Maximization, which enhances the association between the inter-frame feature space and the language space through CMI maximization between them. The details of category-level 6-DoF pose tracking and 3D shape reconstruction are presented in Sec. 4.4 and Sec. 4.5.

4.1 Contrastive Multi-Level Language Alignment

With the multi-level language captions $X_C^{(V,S,O)}$, we are now ready to guide our real pairwise embeddings

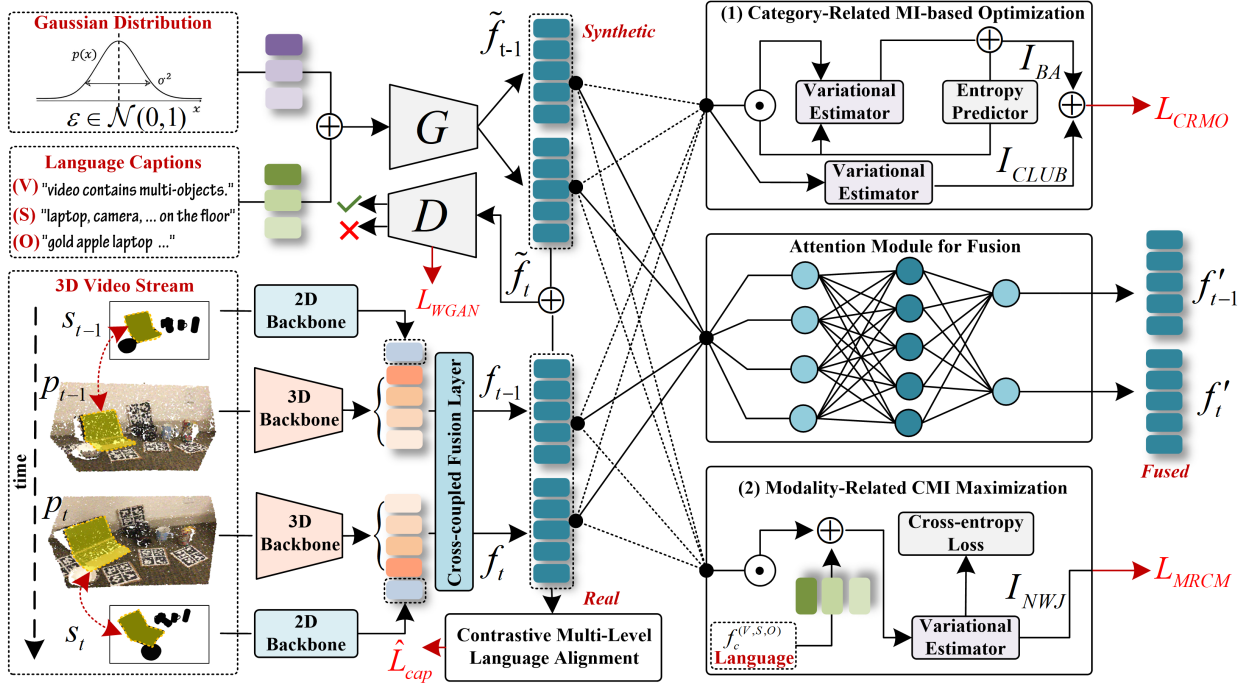


Fig. 2. The overview of the proposed language-4D cross-boosting fusion framework. Our framework takes the 3D video stream with its segmented masks as inputs. The corresponding multi-level language captions are only used during training stage for representation learning, and are not required at inference stage. We firstly process 3D video into the real pairwise embeddings for currently seen classes and then align them with language captions. Besides, we also generate the synthetic pairwise embeddings for unseen classes. These embeddings are cross-boosting fused through two module, i.e., (1) Category-Related MI-based Optimization and (2) Modality-Related CMI Maximization to construct the fused pairwise embeddings for downstream GZSL tasks.

to align with diverse language captions. Since the inter-frame feature pairs f_{t-1} , f_t and language captions X_C are generated independently in their respective spaces, direct alignment could lead to ambiguities. Moreover, unlike conventional language-video 2D image-based modeling and alignment [62], a 3D video stream requires more complex spatial propagation due to its 3D nature. To this end, we introduce a 3D video-language contrastive alignment strategy, which aligns inter-frame feature pairs with their corresponding language instructions from spatial-temporal perspectives. We first obtain the caption embeddings f_c with a pre-trained text encoder F_{text} (ViT-L/14 in [63]) and leverage global average pooling to merge pairwise embeddings f_{t-1} , f_t as follows,

$$f_c = F_{text}(X_c), (\Delta f_{t-1}, \Delta f_t) = Pool(f_{t-1}, f_t). \quad (4)$$

We pool f_{t-1} and f_t separately. To enhance the zero-shot generalization, we incorporate f_c into the inter-frame embeddings f_{t-1} , f_t during training, and no language information is required at inference. We adopt a contrastive loss [64] to align both f_c and Δf across single-level,

$$\mathcal{L}_{cap} = -\frac{1}{2N_{bs}} \sum_{j=t-1}^t \sum_{i=1}^{N_{bs}} \left[\log \frac{\exp(f_c \cdot \Delta f_j^i / \tau)}{\sum_{n=1}^{N_{bs}} \exp(f_c \cdot \Delta f_j^n / \tau)} \right], \quad (5)$$

where N_{bs} is the number of the training batch size and τ is a learnable temperature. With Eq. (4) and Eq. (5), we compute the multi-level caption aligning loss from video-, scene- to object-level. The final alignment loss is then defined as,

$$\hat{\mathcal{L}}_{cap} = \lambda_v \cdot \mathcal{L}_{cap}^{(V)} + \lambda_s \cdot \mathcal{L}_{cap}^{(S)} + \lambda_o \cdot \mathcal{L}_{cap}^{(O)}. \quad (6)$$

4.2 Multi-Level Conditional Feature Generation

Firstly, we proceed to answer "How to obtain effective information that can solve the bias problem?". Inspired by the idea of ZSL [65] using generative adversarial networks (GANs) [66], we aim to alleviate the imbalance between seen/unseen object classes by feeding additional synthetic feature distributions conditioned on multi-level language embeddings (*w.r.t.*, object-, scene-, and video-level) in this section. Thus, we introduce feature generation conditioned on multi-level language descriptions to synthesize the inter-frame pairwise feature embeddings $\{f_{t-1}, \tilde{f}_t\} \in \mathbb{R}^K$ for unseen classes, as shown in Fig. 2. This module consists of a generator G and a discriminator D . The generator G tries to synthesize the pairwise embeddings as follows,

$$(\tilde{f}_{t-1}, \tilde{f}_t) = \mathcal{G}(f_c^{(V,S,O)}, \varepsilon), \quad (7)$$

which takes generated Gaussian feature embeddings and multi-level language embeddings $\{f_c^{(V)}, f_c^{(S)}, f_c^{(O)}\}$ as its inputs. Then, the discriminator D is trained cross-iteratively with the generator G . Our generator is learned by optimizing the following objective,

$$\begin{aligned} \min_G \max_D \mathcal{L}_{WGAN} = & \mathbb{E}[\mathcal{D}((f_{t-1}, f_t), f_c^{(V,S,O)})] \\ & - \mathbb{E}[\mathcal{D}((\tilde{f}_{t-1}, \tilde{f}_t), f_c^{(V,S,O)})] \\ & - \lambda \mathbb{E}[(\|\nabla_{(f_{t-1}, f_t)} \mathcal{D}((\tilde{f}_{t-1}, \tilde{f}_t), f_c^{(V,S,O)})\|_2 - 1)^2] \end{aligned} \quad (8)$$

where $(\hat{f}_{t-1}, \hat{f}_t) = \alpha(f_{t-1}, f_t) + (1 - \alpha)(\tilde{f}_{t-1}, \tilde{f}_t)$ with $\alpha \in U(0, 1)$ and λ is the penalty coefficient. We adopt the adversarial training loss of WGAN in [67] and solve a minmax optimization issue, while D tries to maximize the loss and G tries to minimize it.

4.3 Language-4D Cross-Boosting Fusion

For both real pairwise embeddings derived from a single 3D video stream and synthetic pairwise embeddings generated from language descriptions, although they seem to be independent sequences, there exists a certain correlation between them. In this regard, we hope that the designed model can integrate task-specific information from these embedding vectors into a unified representation, serving as core components for subsequent pose tracking and shape reconstruction tasks. Furthermore, our model should effectively explore potential connections between identical and different modalities, thereby achieving robust zero-shot inference capabilities. Thus, we apply MI-based tractable variational bounds at both the category-related and modality-related levels to optimize unimodal and multimodal fusion.

To make it easier for readers to understand, we begin by discussing the MI tractable variational bounds in detail. Following the basic definition in Eq. (1) and Eq. (2), we construct a variational upper bound by introducing a variational approximation $q(y|x)$ to approximate the marginal distribution $p(y|x)$. Besides, inspired by the special formulation of CLUB in [68], the tractable variational upper bound on MI can be expressed as,

$$I(X; Y) \leq \mathbb{E}_{p(x,y)}[\log q(y|x)] - \mathbb{E}_{p(x)}\mathbb{E}_{p(y)}[\log q(y|x)] \triangleq I_{CLUB}. \quad (9)$$

Likewise, we replace the conditional distribution $p(x|y)$ with a tractable optimization problem over a variational auxiliary distribution approximation $q(x|y)$, yielding the tractable variational lower bound on MI,

$$I(X; Y) = \mathbb{E}_{p(x,y)} \left[\log \frac{q(x|y)}{p(x)} \right] + \mathbb{E}_{p(y)}[D_{KL}[p(x|y)||q(x|y)]] \geq \mathbb{E}_{p(x,y)}[\log q(x|y)] + H(X) \triangleq I_{BA}, \quad (10)$$

where $H(X)$ is the differential entropy of X .

(1) Category-Related MI-based Optimization. Now we proceed to answer the second challenge: “How to increase inter-class distance and decrease intra-class distance at the same time to boost 4D visual-language association for our proposed GZSL task?”. Unlike previous GZSL-based methods [69], [70] that only perform feature learning either in the feature space or in the semantic space independently, our approach aims to leverage the mutual correlation between real and synthetic embeddings, as well as inter-frame embeddings, to facilitate knowledge transfer from seen to unseen classes. This process enhances the discriminability of the fused inter-frame embedding pairs. Concretely, as illustrated in Fig. 2, we minimize mutual information between $f \doteq f_{t-1} \odot f_t$ and $\tilde{f} \doteq \tilde{f}_{t-1} \odot \tilde{f}_t$ (here $\odot$ indicates concatenation) to enhance the model’s discriminability by reducing their intra-class correlation. At each training iteration, we first obtain a batch of data samples $\{(f_i, \tilde{f}_i)\}_{i=1}^{N_{bs}}$. Following [68], we approximate the conditional distribution $q(y|x)$ in Eq. (9) as a Gaussian distribution with parameter θ . We then update

the variational approximation $q_\theta(f_i|\tilde{f}_i)$ by maximizing the log-likelihood,

$$\mathcal{L}_\theta = \frac{1}{N} \sum_{i=1}^N \log q_\theta(f_i|\tilde{f}_i). \quad (11)$$

Once the variational estimator $q_\theta(f_i|\tilde{f}_i)$ is updated, we calculate the MI upper bound between f and $\tilde{f}$ using the formulation from Eq. (9), which is expressed as,

$$I_{CLUB} = \frac{1}{N} \sum_{i=1}^N [\log q_\theta(f_i|\tilde{f}_i) - \frac{1}{N} \sum_{j=1}^N \log q_\theta(f_j|\tilde{f}_i)], \quad (12)$$

where $N = N_{bs}$, and we continuously update $q_\theta(f_i|\tilde{f}_i)$ during the training stage. Following the setup from [68], the variational estimator q_θ is defined as a Gaussian distribution $q_\theta(f_i|\tilde{f}_i) = \mathcal{N}(\tilde{f}_i|\mu(f_i), \sigma^2(f_i) \cdot I)$, where the mean and variance are parameterized by neural networks.

To further increase inter-class distance and eliminate weakly relevant information for downstream tasks, we also aim to maximize the MI lower bound between the inter-frame embedding pairs, *i.e.*, $\{f_{t-1}, f_t\}$ and $\{\tilde{f}_{t-1}, \tilde{f}_t\}$ for seen and unseen classes, respectively. As before, we formulate the conditional distribution $q(x|y)$ in Eq. (10) as multivariate Gaussian distributions, parameterized by two neural networks with θ_1 and θ_2 ,

$$q_\theta(f_{t-1}|f_t) = \mathcal{N}(f_{t-1}|\mu_{\theta_1}(f_t), \sigma_{\theta_2}^2(f_t) \cdot I), \quad (13)$$

$$q_\theta(\tilde{f}_{t-1}|\tilde{f}_t) = \mathcal{N}(\tilde{f}_{t-1}|\mu_{\theta_1}(\tilde{f}_t), \sigma_{\theta_2}^2(\tilde{f}_t) \cdot I), \quad (14)$$

we then update this variational approximation estimator by following likelihood maximization,

$$\mathcal{L}_{\theta_1, \theta_2} = \frac{1}{N} (\sum_{i=1}^N \log q_\theta(f_{t-1}^i|f_t^i) + \sum_{i=1}^N \log q_\theta(\tilde{f}_{t-1}^i|\tilde{f}_t^i)), \quad (15)$$

where $N = N_{bs}$. For the entropy term $H(X)$ in Eq. (10), we solve it using the Gaussian Mixture Model introduced in [71], which can facilitate distribution-based estimation. Ultimately, based on Eq. (10) and Eq. (12), the full loss function for Category-Related MI-based Optimization is given by,

$$\mathcal{L}_{CRMO} = -I_{BA} - I_{CLUB}. \quad (16)$$

Definition 1. (Conditional Mutual Information (CMI)) Like Eq. (1), given three random variables, $x \in X$, $y \in Y$, and $z \in Z$, the Conditional Mutual Information $I(X; Y|Z)$ is defined as,

$$I(X; Y|Z) = \mathbb{E}_{p(x,y,z)} \left[\log \frac{p(x,y,z)}{p(x|z)p(y|z)p(z)} \right] = D_{KL}[p(x,y,z)||p(x|z)p(y|z)p(z)], \quad (17)$$

As suggested by Nguyen *et al.* [72], the variational lower bound on CMI can be represented as follows,

$$I(X; Y|Z) \geq \mathbb{E}_{p(x,y,z)}[\eta(x,y,z)] - e^{-1} \mathbb{E}_{p(x|z)p(y,z)}[\exp \cdot \eta(x,y,z)] \triangleq I_{NWJ}, \quad (18)$$

where $\eta(x,y,z)$ is any optimal function.

(2) Modality-Related CMI Maximization. Here we proceed to answer the third challenge: “How can intra-modal and inter-modal feature fusion be achieved in data-hungry situations?”. To improve GZSL performance, we propose maximizing the CMI lower bound between real and synthetic pairwise embeddings conditioned on language descriptions. This strategy not only enhances the discriminability of feature

embeddings but also strengthens the association between the 3D feature space and the language semantic space. Specifically, given a batch of data $\{(f_i, \tilde{f}_i, f_{ci}^{(V,S,O)})\}_{i=1}^{N_{bs}}$, we first split the data into a training set and a test set. The training set is used to train the estimation network $\eta(x, y, z)$, while the test set is employed to estimate I_{NWJ} . For the optimal function $\eta(x, y, z)$ in Eq. (18), we use a multi-layer neural network, where the final layer applies a sigmoid function to map the output to a real value $\omega \in (0, 1)$. This allows us to train a neural-based classifier to estimate the value of $\eta_\theta(x, y, z)$. We minimize the cross-entropy loss defined as,

$$\mathcal{L}_\theta = -(\sum_{B_{joint}} \log \omega(f_i, \tilde{f}_i, f_{ci}^{(V,S,O)}) + \sum_{B_{prod}} \log(1 - \omega(f_i, \tilde{f}_i, f_{ci}^{(V,S,O)}))), \quad (19)$$

where B_{joint} and B_{prod} represent the joint and the product batches of the current data samples. The output $\omega(*)$ estimates the probability of $(f_i, \tilde{f}_i, f_{ci}^{(V,S,O)})$, thus providing us with the optimal approximation or variational estimator in Eq. (18) as follows,

$$\eta^*(f_i, \tilde{f}_i, f_{ci}^{(V,S,O)}) = 1 + \log \lambda(\omega^*), \quad (20)$$

where $\lambda(\omega^*)$ can be represented as,

$$\lambda(\omega^*) = \frac{\omega^*(f_i, \tilde{f}_i, f_{ci}^{(V,S,O)})}{1 - \omega^*(f_i, \tilde{f}_i, f_{ci}^{(V,S,O)})}. \quad (21)$$

Hence, we can estimate the CMI lower bound based on Eq. (18) as follows,

$$I_{NWJ} = 1 + \sum_{B_{joint}} \log \lambda(\omega^*) - \sum_{B_{prod}} \lambda(\omega^*). \quad (22)$$

In conclusion, the final loss function for Modality-Related CMI Maximization can be expressed as $\mathcal{L}_{MRCM} = -I_{NWJ}$.

We summarize the proposed language-4D cross-boosting fusion algorithm in Alg. 1. Note that the batch-level split is only used to decouple the training of variational estimators from MI/CMI bound estimation, and does not correspond to a dataset-level training/testing split. After achieving the optimization process, we can produce the fused inter-frame embeddings using an attention-based module, as shown in Fig. 2. Using these fused embeddings, we proceed to perform final pose tracking and 3D reconstruction.

4.4 Energy-based Pose Hypothesis

Due to possible ambiguities when inferring the 6-DoF pose change given two 3D views, we introduce an energy-based formulation that can model this uncertainty. Given a pair of fused inter-frame embeddings depicting an arbitrary object, we wish to recover a pose hypothesis distribution over the relative change of pose between two consecutive timesteps $t-1$ and t , as shown in Fig. 3. Inspired by the idea of implicitly representing the distribution using a neural network in recent work [73], we propose to model this conditional distribution of 6-DoF pose hypotheses $P(\Delta \mathcal{P}_t^{(h)} | f'_{t-1}, f'_t)$ as the unnormalized joint log-probability (*w.r.t., energy*),

$$P(\Delta \mathcal{P}_t^{(h)} | f'_{t-1}, f'_t) = \sigma \cdot \exp(F_\eta(\Delta \mathcal{P}_t^{(h)}, f'_{t-1}, f'_t)). \quad (23)$$

Algorithm 1 Language-4D Cross-Boosting Fusion.

Input:

- The dataset of seen classes, $\mathcal{D}^S$;
- Generated multi-level language captions, $X_C^{(V,S,O)}$;
- The number of a batch of sample data, N_{bs} ;

Output:

- Optimized inter-frame pairwise embeddings, f'_{t-1}, f'_t ;
- 1: **for** each training epoch **do**
- 2: Sample a mini-batch from $\mathcal{D}^S$
- 3: Construct real embeddings $\{f_{t-1}, f_t\}$ using Eq. (3)
- 4: Encode multi-level language embeddings $f_{ci}^{(V,S,O)}$
- 5: Align $\{f_{t-1}, f_t\}$ with language captions using Eq. (6)
- 6: Generate synthetic embeddings $\{\tilde{f}_{t-1}, \tilde{f}_t\}$ with Eq. (7)
- 7: Compute log-likelihood in Eq. (11) and Eq. (15)
- 8: Split the mini-batch into:
 - estimator-training subset $\mathcal{B}_{est}$;
 - MI/CMI-estimation subset $\mathcal{B}_{eval}$;
- 9: Compute cross-entropy loss for variational estimator
- 10: Update the parameters of all variational estimators
- 11: Compute MI-based losses $\mathcal{L}_{CRMO}$ and $\mathcal{L}_{MRCM}$
- 12: Update the fusion network by minimizing:
 - $\mathcal{L}_{total} = \hat{\mathcal{L}}_{cap} + \mathcal{L}_{CRMO} + \mathcal{L}_{MRCM}$
- 13: Obtain optimized inter-frame embeddings f'_{t-1}, f'_t
- 14: **end for**
- 15: **return** f'_{t-1}, f'_t

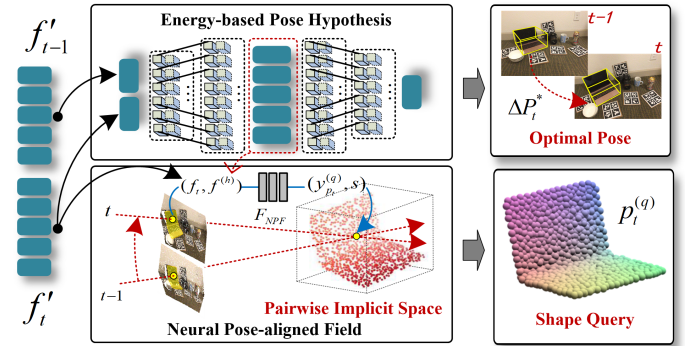


Fig. 3. The illustration of our proposed two downstream GZSL task branches. (a) For 6-DoF pose tracking, we adopt Energy-based Pose Hypothesis to obtain the inter-frame pose change. (b) For 3D shape reconstruction, we introduce Neural Pose-aligned Shape Fields to generate shape query.

To this end, we aim to predict this energy by training a network F_η parameterized by η . Where σ is the constant of integration. From the product rule, this distribution can also be represented as,

$$\begin{aligned} P(\Delta \mathcal{P}_t^{(h)} | f'_{t-1}, f'_t) &= \frac{P(\Delta \mathcal{P}_t^{(h)}, f'_{t-1}, f'_t)}{P(f'_{t-1}, f'_t)} \\ &= \frac{\exp \cdot F_\eta(\Delta \mathcal{P}_t^{(h)}, f'_{t-1}, f'_t)}{\sum_i \exp \cdot F_\eta((\Delta \mathcal{P}_t^{(h)})_i, f'_{t-1}, f'_t)}, \end{aligned} \quad (24)$$

According to Eq. (24), we compute this conditional distribution $\Delta \mathcal{P}_t^{(h)}$ of 6-DoF pose change by sampling multiple homogeneous pose transformation matrices over $\text{SO}(3)$ and $\mathbb{R}^3$, and the number of pose sampling matrices should be large for accurate approximation. Ultimately, we recover the optimal change of pose from frame $t-1$ to t by

optimizing F_η over the 6D space of implicit hypotheses,

$$\begin{aligned}\Delta\mathcal{P}_t^* &= \arg \max_{\Delta\mathcal{P}_t^{(h)} \in \mathbf{SE}(3)} P(\Delta\mathcal{P}_t^{(h)} | f'_{t-1}, f'_t) \\ &= \arg \max_{\Delta\mathcal{P}_t^{(h)} \in \mathbf{SE}(3)} F_\eta(\Delta\mathcal{P}_t^{(h)}, f'_{t-1}, f'_t),\end{aligned}\quad (25)$$

We train the above network F_η by minimizing the negative log-likelihood,

$$\mathcal{L}_{pose} = -\log P(\tilde{\mathcal{P}}_t \cdot \tilde{\mathcal{P}}_{t-1}^{-1} | f'_{t-1}, f'_t), \quad (26)$$

where $\tilde{\mathcal{P}}_t$ and $\tilde{\mathcal{P}}_{t-1}$ are the ground-truth poses of the targeted object in frames $t-1$ and t . $\tilde{\mathcal{P}}_t \cdot \tilde{\mathcal{P}}_{t-1}^{-1}$ denotes the pose change from frame $t-1$ to t .

4.5 Neural Pose-aligned Shape Fields

The key to our approach is learning a pairwise implicit field that learns an inter-frame consistent 3D query shape with pose alignment properties that is related to the current observation. These correspondences are established using NPF defined as a mapping from the pairwise feature f'_t and the pose hypothesis feature $f^{(h)}$ to the corresponding point $y_{p_t}^{(q)}$ in the 3D query shape $p_t^{(q)}$ in the mentioned implicit 3D space, and its signed distance s ,

$$F_{NPF}(f'_t, f^{(h)}; \theta) = (y_{p_t}^{(q)}, s), \quad (27)$$

$$F_{NPF} : \mathbb{R}^K \times \mathbb{R}^K \mapsto \mathbb{R}^3 \times \mathbb{R}, \quad (28)$$

where F_{NPF} denotes a fully connected neural network parameterized by θ . As shown in Fig. 3, the pose hypothesis feature $f^{(h)}$ is extracted from the encoder of the network F_η , as shown in Fig. 3. $y_{p_t}^{(q)}$ denotes the point of the generated shape query. Owing to the fact that the 3D query shape is sampled in the same coordinate frame as the corresponding shape prior, the signed distance s is defined as the distance between $y_{p_t}^{(q)}$ and the surface of the shape prior p_r , which is generated using the Shape Auto-Encoder proposed in [74], and the number of its points is set to N_p . With these fields, we can then calculate the shape regression loss as Eq. (30) and L_1 loss for s similar to [75],

$$\mathcal{L}_s = \frac{1}{N} \sum_i |\text{clamp}(\varphi(\tilde{y}_i), \delta) - \text{clamp}(s, \delta)|, \quad (29)$$

$$\begin{aligned}\mathcal{L}_{shape} &= \frac{1}{N_p} \sum_{x_i \in p_r} \min_{x_j \in p_t^{(q)}} \|x_i - x_j\|_2^2 \\ &+ \frac{1}{N} \sum_{x_i \in p_t^{(q)}} \min_{x_j \in p_r} \|x_j - x_i\|_2^2,\end{aligned}\quad (30)$$

where $\tilde{y}_i$ is the corresponding ground-truth 3D point and δ is a clamping parameter to maintain a metric SDF, as in [76]. We jointly train the energy-based model and the NPF model using the following objective,

$$\eta^*, \theta^* = \arg \min_{\eta, \theta} (\mathcal{L}_{pose} + \mathcal{L}_{shape} + \mathcal{L}_s). \quad (31)$$

5 EXPERIMENTS

In this section, we evaluate our proposed method, named L4D-Track++, for GZSL performance on five benchmarks. The reported pose tracking results on these four unseen datasets include zero-shot category-level pose estimation scenarios.

TABLE 1

Statistical comparison for available datasets used in the experiment. Where "Seen" and "Unseen" respectively indicate the number of seen and unseen categories. Since our method is only trained on NOCS-REAL275, the column of "Unseen" is all based on six seen classes introduced in NOCS-REAL275.

Dataset	Class			Object		Images
	All	Seen	Unseen	Num.	CAD	
NOCS-REAL275 [39]	6	6	0	42	✓	8K
YCB-Video [77]	15	4	11	21	✓	133K
PhoCaL [14]	8	3	5	60	✓	3.9K
HouseCat6D [15]	10	3	7	194	✓	23.5K
Omni6DPose (ROPE) [16]	149	6	143	581	✓	332K

5.1 Experimental Settings

Datasets. We evaluate our proposed GZSL-based method on five available benchmark datasets, *i.e.*, NOCS-REAL275 [39], YCB-Video [77], PhoCaL [14], HouseCat6D [15] and Omni6DPose [16]. The NOCS-REAL275 dataset includes six categories, *i.e.*, *bottle*, *bowl*, *camera*, *laptop*, *can* and *mug*, and comprises 7 training videos and 6 testing videos captured in real-world settings. This data contains 8K images that are collected in 18 different scenes. The YCB-Video dataset comprises both real-world and synthetic images (21 objects), and the real images include 92 videos captured in various scenes using an RGB-D camera. The PhoCaL dataset comprises 60 high-quality 3D models of household objects in eight categories with different photometric complexity. The selected objects include challenging texture-less, occluded, symmetric, reflective and transparent objects. The HouseCat6D dataset encompasses 194 diverse household objects across 10 household categories, comprising high-quality object 3D models including transparent and reflective objects in 41 scenes with broad viewpoint coverage, challenging occlusions and no markers. The Omni6DPose dataset contains real data (ROPE) and simulated data (SOPE); we only verify the results in its real data, which includes 332K images annotated with over 1.5M annotations across 581 instances in 149 categories. The statistics and protocols of these databases are presented in TABLE 1.

Evaluation Metrics. Following the evaluation metrics in [74], [45], we use five aspects of metrics respectively: 1) 3D-IoU, that measures the average precision for various IoU-overlap thresholds, and we use 25%, 50% and 75% for this metric. 2) $a^\circ b$ cm, which measures the average precision of objects for which the error is less than a° for rotation and b cm for translation, and we adopt $5^\circ 2$ cm, $5^\circ 5$ cm, $10^\circ 2$ cm and $10^\circ 5$ cm for this metric. 3) R_{err} and T_{err} , the average error of rotation and translation. 4) ADD (S), the average distance (symmetric) for instance-level pose estimation. 5) CD (Chamfer Distance), which is used for shape reconstruction.

Implementation Details. Our method is only trained on the NOCS-REAL275 dataset and validated on each of the other test sets. The experiments are conducted on an Ubuntu 20.04 system with four NVIDIA RTX A6000 GPUs. We use a batch size of 32 and train the network for 30 epochs with early stopping based on the performance of the model on the held-out validation set, and set the learning rate to be

$1e - 7$. For the penalty coefficient λ and the parameter α in Eq. (8), we fix $\lambda = 10$ and $\alpha = 0.5$. We set the number of hypothesis matrices in Eq. (25) as 5×10^4 . Shape priors are obtained by sampling $N_p = 2048$ points. The trade-off parameters in Eq. (6) are fixed as $\lambda_v = \lambda_s = 0.3, \lambda_o = 0.4$. We compare with different baselines across experiments due to fundamental differences in task definitions and output requirements (pose tracking vs. shape reconstruction). As for the 2D and 3D backbones, we use a CNN-based encoder-decoder framework and the variant of PointNet++ [78], respectively. Their detailed architectures are:

2D-Backbone:

$\text{Resnet}(\text{block}, \text{layer} = [2, 2, 2, 2]) \rightarrow$
 $\text{PSPModule}([512, \text{bins} = (1, 2, 3, 6)]) \rightarrow \text{Dropout}(0.15)$
 $\text{PSPUpsample}([1024, 256, 64, 64, 32]) \rightarrow$
 $\text{FP}(\text{mlp} = [32, 512])$

3D-Backbone:

$\text{SA}(n\text{points} = 2048, \text{radius} = 0.2, \text{mlp}([64, 64, 128])) \rightarrow$
 $\text{SA}(n\text{points} = 2048, \text{radius} = 0.4, \text{mlp}([128, 128, 256])) \rightarrow$
 $\text{SA}(n\text{points} = 2048, \text{radius} = 0.4, \text{mlp}([256, 256, 512])) \rightarrow$
 $\text{BatchNorm}([512]) \rightarrow \text{Dropout}(0.4)$

We use LeakyReLU for each layer in set abstraction (SA), feature propagation (FP), and batch normalization.

5.2 Comparison with State-of-the-Arts

5.2.1 Category-Level Comparison in NOCS-REAL275

To evaluate the pose tracking performance of our method, we conduct category-level 6-DoF pose tracking experiments on the testing set of the NOCS-REAL275 dataset, with results summarized in TABLE 2. Our method is compared against nine state-of-the-art tracking-based baselines, along with their variant models. It is important to note that while these existing methods focus solely on single-pose tracking, our approach additionally addresses dynamic tracking and 3D object reconstruction from video sequences. To ensure a fair comparison, we follow the assumption in [44] that the ground-truth values for pose and size are known in the initial frame. Notably, the evaluation results demonstrate that our approach outperforms both our previous model, L4D-Track [13], and most of the state-of-the-art baselines. Notably, our method achieves significant improvements in several key metrics: $5^\circ 5 \text{ cm}$ (+31.3%), $IoU25$ (+8.4%), R_{err} (+1.3%), and T_{err} (+2.4%) compared to L4D-Track, highlighting the effectiveness of our language-4D cross-boosting fusion module.

We further test the 3D reconstruction capabilities of our approach by comparing it to state-of-the-art RGB-D and depth-based baselines on the NOCS-REAL275 dataset, such as 6D-ViT [79], SPD [43], SGPA [42], and CenterSnap [74]. The results reported in TABLE 3 show that our method, along with L4D-Track, achieves significantly better 3D shape reconstruction performance, with average CD metrics of 0.08 and 0.06, respectively, compared to the current best baseline, ShAPO [80], which achieves a CD of 0.13. This demonstrates that the pairwise implicit 3D field representation proposed in our network significantly enhances the quality of 3D model reconstruction.

TABLE 2
Quantitative comparison of category-level 6-DoF pose tracking on public NOCS-REAL275 dataset. For the metrics of $5^\circ 5 \text{ cm}$ and $IoU25$, the higher the better, and for R_{err} and T_{err} , the lower the better.

Method	Input	Init.	$5^\circ 5 \text{ cm}$	$IoU25$	R_{err}	T_{err}
ICP	Depth	GT.	16.9	47.0	48.1	10.5
Oracle ICP [45]	Depth	GT.	0.65	14.7	40.3	7.7
6-PACK [44]	RGB-D	GT.	28.9	55.4	19.3	3.3
6-PACK w/o temporal	RGB-D	Pert.	22.1	53.6	19.7	3.6
CAPTRA [45]	Depth	Pert.	62.2	64.1	5.9	7.9
CAPTRA +RGB seg.	RGB-D	Pert.	63.6	69.2	6.4	4.2
ICK-Track [46]	RGB-D	GT.	84.4	76.6	4.5	3.1
BundleTrack [47]	RGB-D	GT.	87.4	99.9	2.4	2.1
iCaps [81]	RGB-D	Pert.	31.6	-	9.5	2.3
CATRE [82]	Depth	GT.	57.2	-	6.8	1.2
Corr-Track [50]	Depth	GT.	<u>74.2</u>	<u>94.4</u>	11.0	<u>1.0</u>
Diff-COPE [83]	RGB-D	-	84.3	64.7	-	-
GCE-Pose [84]	Depth	-	67.6	83.3	5.3	4.2
L4D-Track [13]	3D points	GT.	56.2	86.6	<u>5.6</u>	3.3
L4D-Track++ (Ours)	3D points	GT.	87.5	95.0	4.3	0.9

TABLE 3
Quantitative performance comparison of 3D shape reconstruction on public NOCS-REAL275 dataset. All results are evaluated with CD (10^{-2}) and summarized from their original papers.

Method	Bottle	Bowl	Camera	Can	Laptop	Mug	Average
6D-ViT [79]	0.24	0.11	0.61	0.16	0.14	0.11	0.21
CASS [85]	0.17	0.09	0.53	0.18	0.19	0.24	0.23
C3R-Net [86]	0.30	0.10	0.76	0.13	0.13	0.12	0.26
ShAPO [80]	0.10	0.08	0.40	0.07	0.08	0.06	0.13
SPD [43]	0.34	0.12	0.89	0.15	0.29	0.10	0.32
SGPA [42]	0.29	0.09	0.55	0.17	0.16	0.11	0.23
CenterSnap [74]	0.13	0.10	0.43	0.09	0.07	0.06	0.15
GCASP [87]	0.21	0.16	0.11	0.16	0.21	0.29	0.19
L4D-Track [13]	<u>0.09</u>	<u>0.08</u>	<u>0.10</u>	<u>0.07</u>	<u>0.07</u>	<u>0.06</u>	<u>0.08</u>
L4D-Track++(Ours)	0.08	0.07	0.08	0.06	0.06	0.05	0.06

5.2.2 Instance-Level Comparison in YCB-Video

Our method has already shown excellent potential ability in solving category-level tasks for unseen objects with unknown class shifts. However, these results are currently limited to six specific categories; the generalizability of transferable zero-shot learners across different categories and datasets warrants further exploration. To this end, we then conduct zero-shot instance-level 6-DoF pose tracking experiments, training a model on the base classes of the NOCS-REAL275 dataset and testing it on the YCB-Video dataset without any fine-tuning. As depicted in TABLE 4, we select four seen instances and three unseen instances to verify the generalization performance of our method. It can be observed that our method performs well in generalizing to unknown categories and accurately estimates the pose of unseen objects based on seen class knowledge. Specifically, our approach consistently outperforms the latest SoTA instance-level method, se(3)-TrackNet [88], by 20.0% $\sim$ 50.0% in the ADD metric and by 1.0% $\sim$ 2.0% in ADD-S for unseen objects from seen categories, such as "002 master chef can", "006 mustard bottle", and so on. This further illustrates the strong GZSL capability of our approach. Since our model was trained with six

TABLE 4
Quantitative comparison of instance-level 6-DoF pose tracking on public YCB-Video dataset. The results of se(3)-TrackNet are summarized from original paper.

Object	Class	se(3)-TrackNet [88]		L4D-Track++(Ours)	
		ADD	ADD-S	ADD	ADD-S
002 master chef can	Seen	93.8	95.9	97.3	97.9
006 mustard bottle		95.7	97.4	97.0	99.0
024 bowl		39.1	95.6	89.1	96.6
025 mug		91.6	96.9	93.0	97.5
Average		80.1	96.5	93.4	97.8
003 cracker box	Unseen	96.4	97.1	91.7	92.9
011 banana		84.6	95.2	86.3	87.2
019 pitcher base		96.7	97.4	85.7	91.4
Average		92.6	96.6	87.1	89.0

TABLE 5
Quantitative performance comparison of 3D shape reconstruction on public YCB-Video dataset. We evaluate performance with CD (10^{-2}) using their public checkpoints or codes.

Object	Class	SGPA [42]	SPD [43]	Center-Snap [74]	L4D-Track [13]	Ours
002 master chef can	Seen	0.16	0.15	0.18	<u>0.14</u>	0.12
006 mustard bottle		0.31	0.55	<u>0.14</u>	0.20	0.18
024 bowl		0.10	0.13	0.13	<u>0.09</u>	0.08
025 mug		0.14	0.12	0.07	0.11	<u>0.10</u>
003 cracker box	Unseen	0.23	0.16	0.17	<u>0.16</u>	0.14
011 banana		0.21	0.34	<u>0.20</u>	0.25	0.18
019 pitcher base		0.30	0.27	0.22	<u>0.19</u>	0.17
Average	#	0.21	0.25	<u>0.16</u>	<u>0.16</u>	0.14

specific categories from the NOCS-REAL275 dataset, its performance on other unseen categories (*e.g.*, "011 banana") is relatively weaker than the current best method, se(3)-TrackNet. Furthermore, we present detailed quantitative comparisons of 3D shape reconstruction on the YCB-Video dataset depicted in TABLE 5, where we compare with three representative baselines, including SGPA [42], SPD [43], CenterSnap [74], and our previous work, L4D-Track [13]. Likewise, we select seven categories, including four seen classes and three unseen classes. As depicted in TABLE 5, we obtain average CD metrics of 0.16 and 0.14, respectively. We outperform SoTA baselines, indicating the superiority for GZSL 3D shape reconstruction.

5.2.3 GZSL Performance Evaluation in PhoCaL

To quantitatively evaluate the GZSL performance of our method on household objects with different photometric complexity, including challenging texture-less, occluded, symmetric, reflective and transparent instances, we also perform a comprehensive comparison using our trained model. We apply it to both seen and unseen classes on the public PhoCaL [14] dataset and compare its performance against several state-of-the-art baselines, *e.g.*, NOCS [39] and CPS [89]. The detailed results are shown in TABLE 6 and TABLE 8. For the category-level pose estimation/tracking task, as shown in TABLE 6, our GZSL-based method achieves significant improvements in terms of average precision for various IoU-overlap thresholds. Specifically,

we yield an average accuracy of 83.90% for seen classes and 30.04% for unseen classes. Notably, while our method outperforms others in both seen and unseen classes, it is especially effective in handling unseen classes. This is in contrast to the NOCS and CPS methods, which are estimation-based and perform well on seen classes but fail to generalize to unseen classes. Our tracking-based approach consistently exhibits strong zero-shot ability, showcasing the effectiveness of our GZSL model. For the 3D shape reconstruction task, as illustrated in the left part of TABLE 8, we compare our method with several competitive baselines, including SGPA [42], SPD [43], and CenterSnap [74]. Our method achieves impressive results, with a 0.23 higher average accuracy in the Chamfer Distance (CD) metric compared to other baselines. Moreover, our method shows particularly strong performance on unseen classes, with improvements of 0.30 for "Remote" and 0.20 for "Box". This performance boost can be attributed to the cross-boosting fusion module, which leverages semantic information from language captions, enabling better transfer of knowledge to unseen classes. In contrast, other methods struggle to transfer meaningful information from seen to unseen classes, especially when the unseen categories are dissimilar to the seen ones. The superiority of our method is also evident in the visualizations for unseen categories, as shown in Fig. 4(a), further highlighting the effectiveness of our approach in both pose tracking and 3D shape reconstruction.

5.2.4 GZSL Performance Evaluation in HouseCat6D

To further evaluate the effectiveness of our proposed GZSL-based method, we conduct experiments on the HouseCat6D dataset [15], which covers a wide range of scenes with diverse household object categories and varying photometric complexities. This dataset includes large viewpoint distributions, providing a robust test for generalization. We report the 3D IoU results with 25% and 50% thresholds for category-level pose estimation/tracking in TABLE 7, and the 3D shape reconstruction results, measured by Chamfer Distance (CD), in the right part of TABLE 8. As shown in TABLE 7, our GZSL-based method outperforms several state-of-the-art pose estimation baselines, including NOCS [39], FS-Net [90], GPV-Pose [91], VI-Net [92], and SecondPose [93], achieving the best results for three seen classes and surpassing the others in most unseen classes on the HouseCat6D dataset. Notably, our method achieves an average 3D IoU improvement of 97.4% and 67.0% over the baseline methods. This is particularly significant because our method was trained solely on the NOCS-REAL275 dataset, while all other baselines were trained on the HouseCat6D dataset, highlighting the strong zero-shot generalization ability of our approach. In addition to pose tracking, we also evaluate the GZSL performance for 3D shape reconstruction, as shown in TABLE 8. Our model not only achieves the best results for seen classes but also significantly outperforms all other methods across all unseen classes. This demonstrates the effectiveness of our language-4D cross-boosting fusion design in improving the generalization ability for shape reconstruction tasks. Finally, we provide visual comparisons of pose tracking and shape reconstruction for unseen classes in the HouseCat6D

TABLE 6

Quantitative performance comparison of category-level 6-DoF pose tracking on public PhoCal dataset. All compared results are summarized from the original paper [14]. The best and the second best results for different metrics are highlighted in **bold** and underlined.

Method	Average <i>IoU25/IoU50</i>	Seen Class			Unseen Class				
		Bottle	Can	Cup (Mug)	Box	Remote	Teapot	Cutlery	Glassware
NOCS [39]	43.34 / 13.91	<u>91.17</u> / 0.65	<u>85.44</u> / <u>23.01</u>	51.83 / 1.48	16.10 / 0.01	93.26 / 86.05	0.00 / 0.00	4.89 / 0.01	4.00 / 0.06
CPS [89]	<u>61.30</u> / <u>28.69</u>	80.08 / <u>40.30</u>	68.96 / 6.69	<u>81.60</u> / 70.24	<u>31.68</u> / <u>28.18</u>	86.30 / <u>37.08</u>	<u>67.43</u> / <u>4.31</u>	<u>44.00</u> / 24.95	<u>30.33</u> / <u>17.74</u>
Ours	83.90 / 30.04	98.00 / 82.13	97.03 / 52.36	82.14 / <u>24.15</u>	97.23 / 41.96	<u>86.90</u> / 20.00	67.91 / 5.24	78.45 / <u>18.19</u>	98.25 / 40.00

TABLE 7

Quantitative performance comparison of category-level 6-DoF pose tracking on public HouseCat6D dataset. All compared results are summarized from the original paper [15]. The best and the second best results for different metrics are highlighted in **bold** and underlined.

Method	Average <i>IoU25/IoU50</i>	Seen Class			Unseen Class						
		Bottle	Can	Cup (Mug)	Box	Remote	Teapot	Cutlery	Glass	Tube	Shoe
NOCS [39]	50.0 / 21.2	41.9 / 5.0	81.9 / 62.4	68.8 / 2.0	43.3 / 6.5	<u>81.8</u> / 59.8	24.3 / 0.1	14.7 / 6.0	95.4 / 49.6	21.0 / 4.6	26.4 / 16.5
FS-Net [90]	74.9 / 48.0	65.3 / 45.0	98.3 / 73.8	96.4 / 68.1	31.7 / 1.2	<u>65.6</u> / 46.8	69.9 / 59.8	71.0 / 51.6	99.4 / 32.4	79.7 / 46.0	71.4 / 55.4
GPV-Pose [91]	74.9 / 50.7	66.8 / 45.6	98.6 / 75.2	96.7 / 69.0	31.4 / 1.1	65.7 / 46.9	75.4 / <u>61.6</u>	70.9 / 52.0	99.6 / 62.7	76.9 / <u>42.4</u>	67.4 / 50.2
VI-Net [92]	80.7 / 56.4	90.6 / 79.6	<u>99.0</u> / 67.0	96.7 / 72.1	44.8 / 12.7	54.9 / 17.1	52.6 / 47.3	89.2 / 76.4	99.1 / 93.7	94.9 / 36.0	85.2 / <u>62.4</u>
SecondPose [93]	<u>83.7</u> / <u>66.1</u>	<u>94.5</u> / <u>79.8</u>	98.5 / 93.2	99.8 / 82.9	<u>54.5</u> / <u>23.7</u>	53.6 / 35.4	<u>81.0</u> / 71.0	<u>93.5</u> / <u>74.4</u>	99.3 / <u>92.5</u>	75.6 / 35.6	<u>86.9</u> / 73.0
Ours	97.4 / 67.0	99.3 / 94.3	99.9 / <u>92.3</u>	<u>98.5</u> / <u>79.6</u>	99.0 / 92.3	98.0 / <u>44.9</u>	91.1 / 20.0	97.2 / 60.0	99.1 / 82.4	<u>85.4</u> / 20.0	96.7 / 40.0

TABLE 8

Quantitative performance comparison of 3D shape reconstruction on both public PhoCal and HouseCat6D dataset. We evaluate performance with CD (10^{-2}) using their public checkpoints or codes and the shape priors for unseen classes are constructed based on SPD.

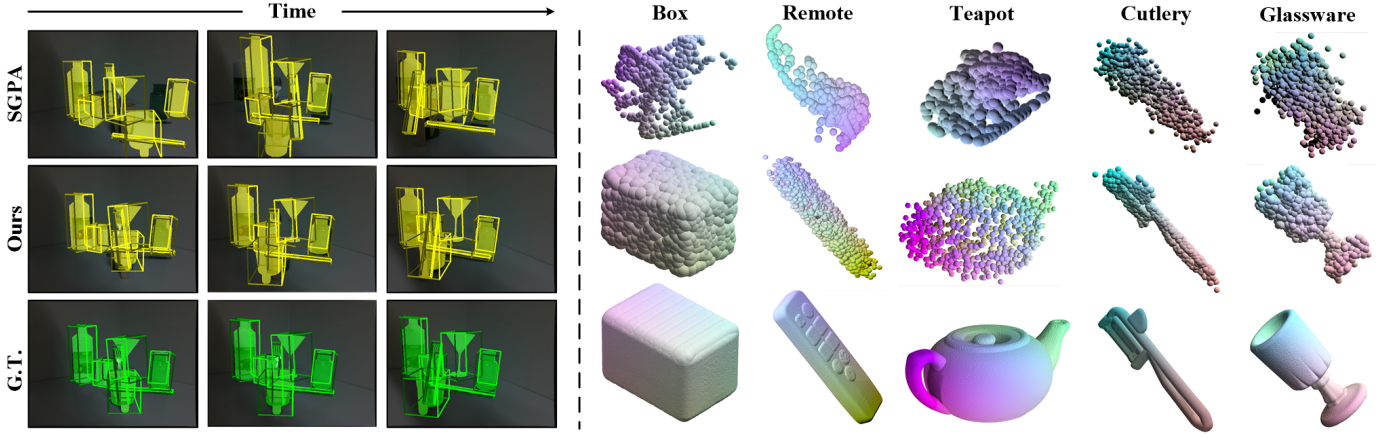
Dataset		PhoCal Dataset [14]				HouseCat6D Dataset [15]			
Object	Class	SGPA [42]	SPD [43]	CenterSnap [74]	Ours	SGPA [42]	SPD [43]	CenterSnap [74]	Ours
Bottle	Seen	0.23	<u>0.14</u>	0.19	0.12	0.21	<u>0.20</u>	0.25	0.12
Can		0.28	<u>0.19</u>	0.29	0.16	0.30	0.33	<u>0.27</u>	0.25
Cup (Mug)		0.26	<u>0.19</u>	0.32	0.17	0.25	0.28	<u>0.21</u>	0.19
Box	Unseen	0.45	0.68	<u>0.42</u>	0.20	<u>0.62</u>	0.73	0.71	0.23
Remote		0.62	0.68	<u>0.57</u>	0.30	<u>0.57</u>	0.75	0.89	0.33
Teapot		0.50	0.49	<u>0.35</u>	0.30	0.45	<u>0.38</u>	0.54	0.35
Cutlery		0.69	0.62	<u>0.59</u>	0.41	<u>0.37</u>	0.39	0.44	0.32
Glass (Glassware)		<u>0.39</u>	0.48	0.42	0.30	0.50	0.52	<u>0.49</u>	0.44
Tube		-	-	-	-	<u>0.29</u>	0.31	0.33	0.26
Shoe		-	-	-	-	<u>0.35</u>	<u>0.35</u>	0.40	0.34
Average	#	0.43	0.43	<u>0.39</u>	0.23	<u>0.39</u>	0.42	0.45	0.28

dataset, as shown in Fig. 4(b). Notably, SGPA [42] produces incomplete shapes due to its reliance on shape priors, further emphasizing the advantages of our approach in handling unseen object categories.

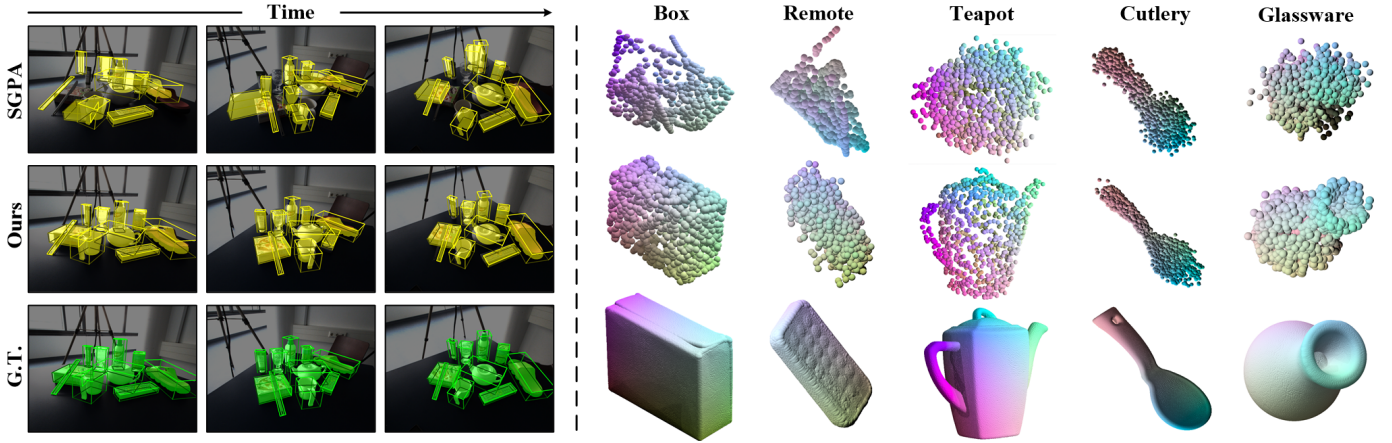
5.2.5 GZSL Performance Evaluation in Omni6DPose

To evaluate the fine-grained GZSL performance of our proposed method, we conduct experiments on the large-scale Omni6DPose dataset [16], which consists of 149 instance categories. For these experiments, we focus exclusively on the real-world test set. We also include a comparison with our previous work, L4D-Track [13]. The detailed experimental results are presented in TABLE 9 and TABLE 10. For category-level 6-DoF pose tracking, we adopt BundleTrack [47], CATRE [82], and GenPose (with its improved model) as baselines, as done in [16]. The results, averaged across all 149 categories, are shown in TABLE 9. Our GZSL-based method achieves significant improvements over the state-of-the-art, with relative gains of 56.1%, 63.5%, 6.3%, and 2.5% in key tracking metrics.

Moreover, compared to our previous model, L4D-Track, we observe further improvements of 9.3%, 3.3%, 1.4%, and 0.7% in pose tracking accuracy. These results demonstrate that our method performs effectively in both the standard ZSL and the more challenging GZSL setting. Regarding 3D shape reconstruction, we evaluate performance on both seen and unseen classes, selecting six seen classes and four unseen classes, including Dish, Teapot, Book, and Vase, as shown in TABLE 10. Our method consistently outperforms the baselines on all seen classes and achieves significant improvements in the unseen classes. The competitive results under the GZSL setting further validate the effectiveness of our language-4D cross-boosting fusion design. Our method addresses robust cross-category generalization rather than optimizing performance for specific seen categories. While the absolute performance may be lower in some cases, the proposed method consistently improves generalization to unseen categories compared to baselines, demonstrating its effectiveness under the GZSL setting.



(a) The comparison results on public PhoCal dataset.



(b) The comparison results on public HouseCat6D dataset.

Fig. 4. Visual comparison on PhoCal and HouseCat6D datasets. **Left**: the qualitative comparison of 6-DoF pose tracking. For a clearer comparison, we have expressed the results in RGB images. Yellow and green represent the results from SGPA [42], ours and ground-truth label, respectively. **Right**: the output of final reconstructed 3D shape, that are rendered from the same viewpoint.

TABLE 9

Quantitative comparison of category-level 6-DoF pose tracking on public Omni6DPose dataset. For $5^\circ 5cm$ and $mIoU$, the higher the better, and for R_{err} and T_{err} , the lower the better. The results are averaged over all 149 categories and summarized from [16]. GT. Pert. denotes that a perturbed ground truth pose is utilized as the initial pose. $mIoU$ indicates Mean Intersection over Union, representing the average 3D overlap between the ground truth and the predicted bounding boxes.

Method	Input	Init.	$5^\circ 5cm$	$mIoU$	R_{err}	T_{err}
BundleTrack [47]	RGBD	GT.	1.3	3.9	46.9	23.5
CATRE [82]	Depth	GT.Pert.	15.9	55.4	21.3	2.6
GenPose [94]	Depth	GT.Pert.	13.3	-	19.3	1.2
Genpose++ [16]	RGBD	GT.Pert.	15.9	53.4	17.6	1.2
RFMPose [95]	RGBD	-	16.1	54.1	15.9	1.2
L4D-Track	3D points	GT.Pert.	<u>46.8</u>	<u>60.2</u>	<u>7.7</u>	3.2
L4D-Track++ (Ours)	3D points	GT.Pert.	56.1	63.5	6.3	<u>2.5</u>

TABLE 10

Quantitative comparison of 3D shape reconstruction on public Omni6DPose dataset. We evaluate performance with CD (10^{-2}) using their public checkpoints or codes and the shape priors for unseen classes are constructed based on SPD.

Object	Class	SGPA [42]	SPD [43]	CenterSnap [74]	L4D-Track [13]	L4D-Track++ (Ours)
Bottle	Seen	0.48	0.31	<u>0.27</u>	0.28	0.26
bowl		<u>0.22</u>	0.39	0.44	0.25	0.20
Mug		0.47	0.26	0.27	<u>0.25</u>	0.23
Can		0.35	<u>0.28</u>	0.29	0.30	0.21
Camera		0.37	0.26	0.31	<u>0.25</u>	0.23
Laptop	Unseen	0.35	0.30	0.40	<u>0.22</u>	0.20
Dish		0.52	0.64	0.68	<u>0.30</u>	0.28
Teapot		0.41	<u>0.31</u>	0.37	0.35	0.30
Book		0.54	0.45	0.50	<u>0.36</u>	0.32
Vase		0.61	0.79	0.71	<u>0.29</u>	0.28
Average	#	0.43	0.40	0.42	<u>0.29</u>	0.25

5.3 Ablation Studies and Additional Analysis

In this section, we conduct a series of ablation experiments and additional analyses to evaluate the contribution and functionality of each component in our proposed method.

5.3.1 Effect of Language-4D Cross-Boosting Fusion

The proposed language-4D cross-boosting fusion method consists of three key components: Category-Related MI-

TABLE 11

Ablation study of language-4D cross-boosting fusion. Where I_{BA} , I_{CLUB} , I_{NWJ} and Fusion denote the each component of our proposed language-4D cross-boosting fusion module.

Method	NOCS-REAL275 Dataset						PhoCal Dataset		
I_{BA} I_{CLUB} I_{NWJ} Fusion	$5^\circ 5cm$	$IoU25$	R_{err}	T_{err}	CD		$IoU25$	$IoU50$	CD
✗ ✗ ✗ ✗	56.2	86.6	5.6	3.3	0.08		60.1	12.4	0.57
✓ ✗ ✗ ✗	62.4	90.7	4.8	2.2	0.07		61.5	15.3	0.40
✓ ✓ ✗ ✗	87.4	94.9	4.4	1.0	0.06		62.5	13.9	0.49
✓ ✓ ✓ ✗	87.0	94.8	4.5	1.3	0.06		83.8	29.7	0.25
Ours	87.5	95.0	4.3	0.9	0.06		83.9	30.0	0.23

TABLE 12

Ablation study of Multi-Level Conditional Feature Generation. Where "object-level", "scene-level" and "video-level" denote the corresponding language caption embedding, and "class semantic" represents we only use the class-level categorical semantic embedding for synthetic feature generation.

Embedding Inputs	NOCS-REAL275 Dataset						PhoCal Dataset		
	$5^\circ 5cm$	$IoU25$	R_{err}	T_{err}	CD		$IoU25$	$IoU50$	CD
only w/ object-level	87.3	94.1	4.2	1.0	0.06		80.9	26.7	0.42
only w/ scene-level	86.8	94.3	4.3	1.1	0.07		79.4	25.8	0.39
only w/ video-level	87.1	94.0	4.0	1.0	0.06		79.9	28.9	0.38
only w/ class semantic	58.4	87.0	5.5	3.2	0.08		62.2	10.3	0.65
Ours	87.5	95.0	4.3	0.9	0.06		83.9	30.0	0.23

based Optimization (I_{BA} , I_{CLUB}), Modality-Related CMI Maximization (I_{NWJ}), and the Attention Module for Fusion (Fusion). This section evaluates the importance of each component through an ablation study, presented in TABLE 11. The experiment assesses pose tracking and reconstruction performance for both seen and unseen classes across two public datasets. Notably, the model without any components corresponds to our previous work, L4D-Track, as shown in row 1 of the table. As indicated in TABLE 11, the method incorporating both Category-Related MI-based Optimization and Modality-Related CMI Maximization achieves the best performance for seen classes. Moreover, it demonstrates superior GZSL generalization to unseen classes. These results suggest that Category-Related MI-based Optimization improves the discriminability of fused inter-frame embeddings by increasing inter-class distance and decreasing intra-class distance. Similarly, Modality-Related CMI Maximization strengthens the association between the feature space and the language semantic space, both of which are crucial for enhancing GZSL generalization in downstream tasks.

5.3.2 Impact of Synthetic Pairwise Feature Generation

To gain deeper insights into the synthetic pairwise embeddings for unseen classes, we conduct ablation studies to evaluate the impact of our proposed Multi-Level Conditional Feature Generation module. This module addresses the imbalance between seen and unseen object classes by conditioning on multi-level language embeddings. The results of these experiments, shown in TABLE 12, highlight the influence of different combinations of language embeddings. Specifically, we verify each

TABLE 13

Ablation study of Contrastive Multi-Level Language Alignment. Where $L_{cap}^{(V)}$, $L_{cap}^{(S)}$ and $L_{cap}^{(O)}$ denote single-level cross-modality alignment loss.

Loss Terms	NOCS-REAL275 Dataset						PhoCal Dataset		
$L_{cap}^{(V)}$ $L_{cap}^{(S)}$ $L_{cap}^{(O)}$	$5^\circ 5cm$	$IoU25$	R_{err}	T_{err}	CD		$IoU25$	$IoU50$	CD
✗ ✗ ✗	86.4	93.9	5.0	1.3	0.10		78.4	23.8	0.28
✓ ✗ ✗	86.8	94.2	4.5	1.4	0.08		79.3	25.9	0.30
✓ ✓ ✗	87.0	94.7	4.8	1.1	0.12		80.0	26.0	0.33
Ours	87.5	95.0	4.3	0.9	0.06		83.9	30.0	0.23

TABLE 14

Ablation study on the robustness to pose noises. Init. $\times m$ means adding m times train-time errors in pose initialization. All $\times m$ means adding m times errors to all estimated poses in every previous frames.

Dataset	Metric	Orig.	Init. $\times 1$	Init. $\times 2$	All $\times 1$	All $\times 2$
NOCS-REAL275	$5^\circ 5cm \uparrow$	87.5	86.0	84.9	85.5	83.7
	$IoU25 \uparrow$	95.0	85.9	88.8	88.7	87.1
	$R_{err} \downarrow$	4.3	5.8	6.4	5.5	5.8
	$T_{err} \downarrow$	0.9	1.2	1.3	1.0	1.5
PhoCal	$IoU25 \uparrow$	83.9	77.7	75.6	78.0	76.9
	$IoU50 \uparrow$	30.0	23.0	22.4	24.7	22.1

configuration using a single language caption embedding in turn, as well as class-level category semantic embeddings. Our method consistently achieves higher accuracy in both pose tracking and shape reconstruction compared to other approaches. These demonstrate that multi-level language embeddings are effective in generating meaningful representations for unseen classes, and mitigate the bias introduced by class imbalances.

5.3.3 Influence of Contrastive Language Alignment

To evaluate the effectiveness of the language contrastive alignment module, we conduct an ablation study by reducing the contrastive loss from the video-level to the object-level, as shown in TABLE 13. The results clearly demonstrate that all ablated versions exhibit poorer performance on both datasets when the corresponding loss is removed. This highlights the importance of supervision from multi-level captions and provides indirect evidence that our designed language-to-4D association module contributes to the overall performance improvement of the model.

5.3.4 Tracking Robustness to Pose Noises

Since our method relies on the pose from the previous frame or initial pose, we conduct an ablation study to evaluate its robustness against pose noise and its impact on pose accuracy. As shown in TABLE 14, we progressively increase initial pose error by one or two times to assess robustness to pose initialization. Additionally, we introduce pose errors at one or two time intervals for each subsequent frame to examine robustness to tracking pose errors. The results demonstrate that our method is robust to noise in both the initial and previous poses.

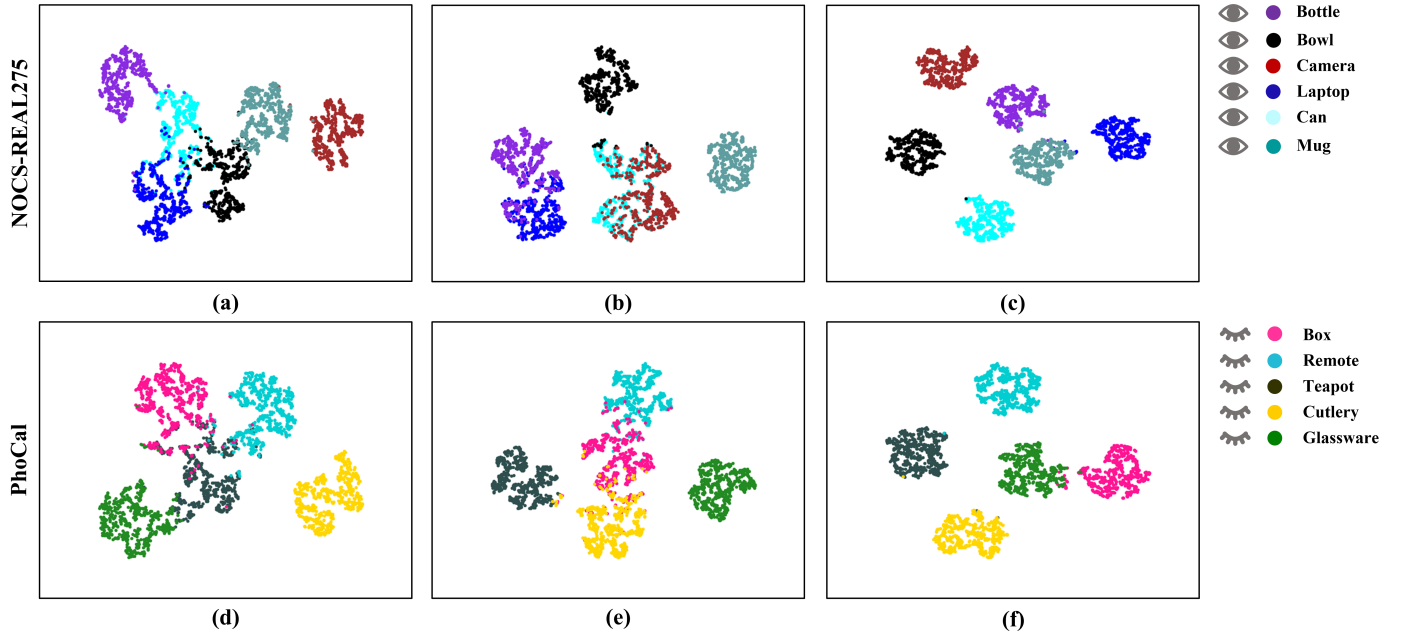


Fig. 5. Visualization comparison of t-SNE for pairwise feature embeddings. **Figure (a)-(c)** are the representation distributions of real/synthetic pairwise embeddings and fused pairwise embeddings for all seen classes (including Bottle, Bowl, Camera, Laptop, Can and Mug) from NOCS-REAL275 dataset. **Figure (d)-(f)** show the representation distributions of real/synthetic pairwise embeddings and fused pairwise embeddings for unseen classes (*i.e.*, Box, Remote, Teapot, Cutlery and Glassware) from PhoCal dataset.

5.3.5 t-SNE Quantitative Visualisation Analysis

To further demonstrate the effectiveness of the proposed language-4D cross-boosting fusion, we perform t-SNE visualization for both seen classes from the NOCS-REAL275 dataset and unseen classes from the PhoCal dataset, as shown in Fig. 5. Specifically, Fig. 5(a) and (b) present the visualization of real pairwise embeddings using the pre-trained 2D/3D backbones from our L4D-Track model, and synthetic pairwise embeddings derived from language captions, respectively. Fig. 5(c) shows the final fused pairwise embeddings for seen classes on NOCS-REAL275, while Fig. 5(d)–(f) depict the corresponding embeddings for unseen classes on PhoCal. As observed, the embeddings produced by previous baselines tend to crowd together, with significant overlap between certain categories (*e.g.*, Bowl and Mug). The synthetic pairwise features show an improvement in clustering, though some overlap remains, as seen in Fig. 5(b). In contrast, the fused embeddings exhibit clearer clustering for different seen classes, and the embeddings for unseen classes from the PhoCal dataset are also well clustered. Specifically, our GZSL-based model produces pairwise embeddings that significantly enhance clustering for both seen and unseen classes. This result demonstrates that the proposed fusion module effectively captures the relationship between 4D features and language captions, maximizing inter-class distance while minimizing intra-class distance. As a result, it learns and transfers language-related knowledge to the unseen domain, benefiting downstream tasks such as tracking and reconstruction.

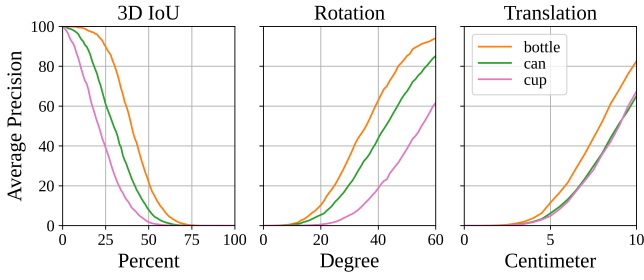
5.3.6 Extra mAP Qualitative Comparison

We report the category-specific average precision for 6-DoF pose tracking on the public PhoCal and HouseCat6D datasets, evaluated under different thresholds using 3D

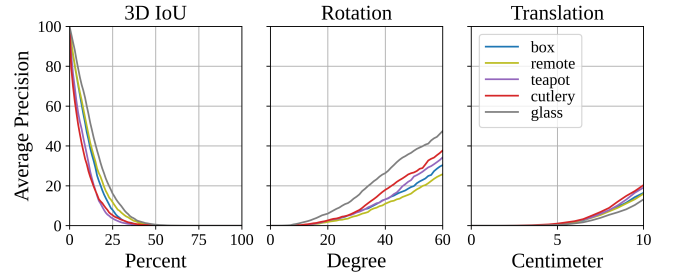
IoU, as well as rotation and translation errors. We compare these results with the current state-of-the-art category-level baseline, SGPA [42]. To better analyze the GZSL performance of our method, we separately evaluate the unseen and seen categories. The comparisons are presented in Fig. 6, where the results demonstrate that, across all evaluation metrics, our method outperforms SGPA in every category of both datasets. Notably, compared to the category-specific model in SGPA, our approach leverages the proposed language-4D Cross-Boosting module, enabling zero-shot inference for both seen and unseen classes. In contrast, SGPA can only effectively handle pose tracking for seen classes (*e.g.*, bottle, can, and cup), while its performance on unseen classes is significantly poor, often failing to provide pose estimation results. These trends are consistent across both public datasets, further validating the superiority of our method in GZSL tasks.

5.3.7 Computational Complexity and Runtime Analysis

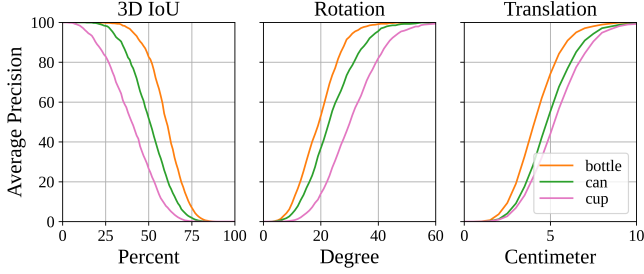
We also report the runtime and computational complexity analysis in TABLE 16. Although the proposed framework involves multiple components during training, all auxiliary modules, including synthetic conditional-based embedding generation, MI/CMI estimators, and language supervision, are used only in the training stage and are removed at inference time. Consequently, the inference pipeline consists solely of the frozen 2D/3D backbone and a lightweight fusion head, whose computational cost is comparable to that of prior methods. As shown in TABLE 16, the full model achieves comparable or even faster inference speed than existing approaches, demonstrating that the increased training complexity does not hinder real-world deployment. This design allows the proposed framework to benefit from rich supervision during training while maintaining an efficient and deployment-friendly inference pipeline.



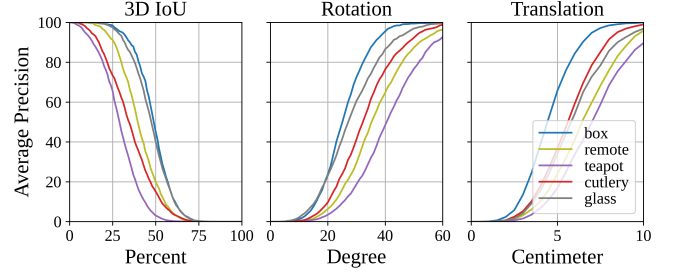
(a) SGPA [42] results for seen classes on PhoCal dataset.



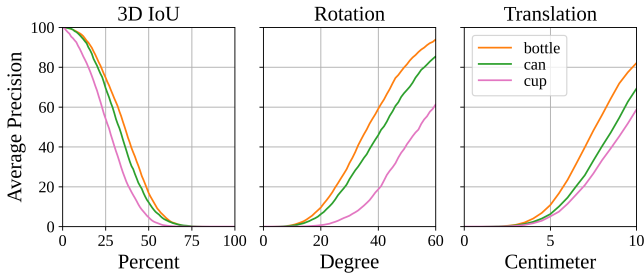
(b) SGPA [42] results for unseen classes on PhoCal dataset.



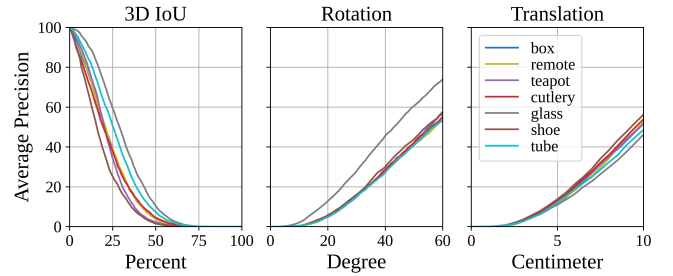
(c) Our results for seen classes on PhoCal dataset.



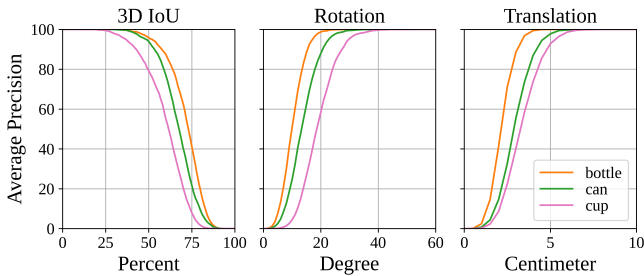
(d) Our results for unseen classes on PhoCal dataset.



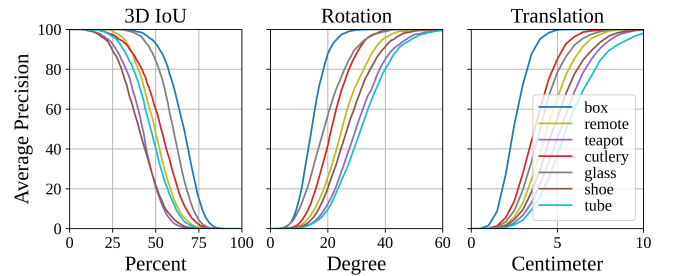
(e) SGPA [42] results for seen classes on HouseCat6D dataset.



(f) SGPA [42] results for unseen classes on HouseCat6D dataset.



(g) Our results for seen classes on HouseCat6D dataset.



(h) Our results for unseen classes on HouseCat6D dataset.

Fig. 6. Comparison of mAP on two public datasets. We compare our method with typical baseline (SGPA [42]) under the metric of 3D IoU, rotation and translation for seen classes (*i.e.*, **Figure (a), (c), (e) and (g)**) and unseen classes (*i.e.*, **Figure (b), (d), (f) and (h)**), respectively.

TABLE 15

Training cost analysis. We report the average training time per iteration, GPU memory usage, and relative overhead compared with the backbone-only baseline on NVIDIA RTX A6000 GPU.

Method	Time/Iter (ms)	GPU Memory (GB)	Training Overhead	Inference Overhead
Backbone-only	125	23.5	-	-
Ours (Full Model)	142	26.0	+13.6%	<5%

TABLE 16

Comparison of computational complexity and runtime analysis. Note that all results are from models using their public checkpoints, and all methods are evaluated on the same GPU for fair comparison.

Metric	SPD [43]	SGPA [42]	Ours w/o cross fusion	Ours w/o attention fusion	Ours (full)
Params (M)	18.30	23.31	19.54	18.71	21.78
Inference (ms)	65.66	70.84	50.11	50.89	53.26

5.3.8 Training Cost and GPU Memory Analysis

TABLE 15 summarizes the computational cost and GPU memory overhead introduced by the proposed modules. Compared with the backbone-only baseline, the full model increases the training time per iteration by approximately 11.9% and GPU memory usage by approximately 13.6%. Overhead is a relative metric defined with respect to the backbone-only baseline and thus is not applicable to the baseline itself. It is worth noting that several components, such as MI-based optimization and language-conditional feature generation, are only used during training. As a result, the inference-time overhead remains negligible (less than 5%), making the proposed method suitable for real-world deployment.

6 CONCLUSION

Providing full 6-DoF pose and 3D shape information for observable target objects remains a significant challenge in robotics and computer vision applications. In this paper, we introduce a simple yet effective multi-modal fusion strategy, called language-4D cross-boosting fusion, which enables GZSL-based simultaneous category-level pose tracking and 3D shape reconstruction for both seen and unseen classes. This is achieved by facilitating the exploitation of language-to-4D modality interactions. Specifically, the proposed language-4D cross-boosting fusion consists of two key components: Category-Related MI-based Optimization and Modality-Related CMI Maximization. These components can serve as a flexible plug-in framework, adaptable to a variety of inter-modal or intra-modal fusion methods. Quantitative and qualitative experimental results demonstrate that our GZSL-based method not only compares favorably with existing approaches but also surpasses state-of-the-art methods in performance.

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